SİNYALLER VE SİSTEMLER

Sinyallerin Sınıflandırılması (Değişken Sayısı)

1- Bir boyutlu (1D): Sadece tek bir boyuta sahip sinyaller.

Örnek: Ses sinyali, biyomedikal sinyaller, sıcaklık sinyali

2- İki boyutlu (2D): Eğer sinyal fonksiyonu iki adet değişken içeriyorsa bu iki boyutlu sinyaldir.

Örnek: resim sinyali

3- Üç boyutlu (3D): Eğer sinyal fonksiyonu üç adet değişken içeriyorsa bu üç boyutlu sinyaldir.

Örnek: video sinyali

Sinyallerin Niteliğine Göre Sınıflandırılması

1- Sürekli Zaman Sinyali (CTS) veya Analog Sinyal

Sinyallerin değerleri zamana göre sürekli bir şekilde değişiyorsa buna sürekli zaman sinyali denir.

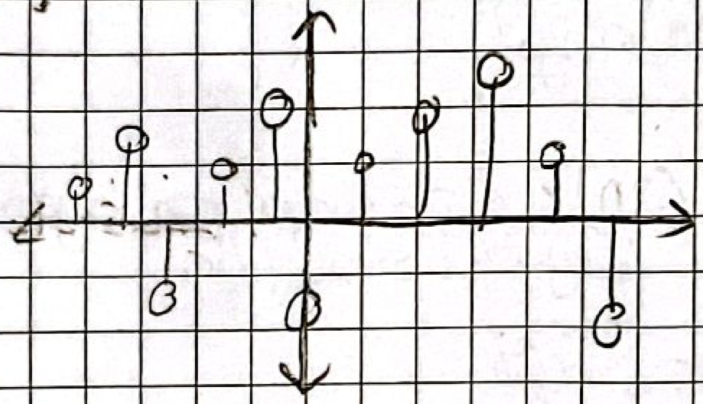
Sonsuz değerler kümesi içerir.

2- Dijital Sinyal

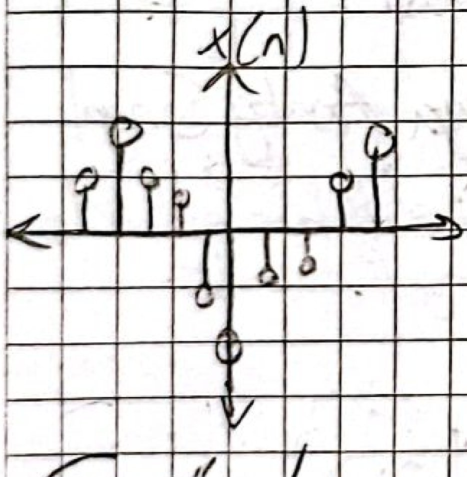
Sinyal yalnızca iki değer içeriyorsa, buna dijital sinyal denir. [0,1]

3- Ayrık Zamanlı Sinyal

Sinyal zamana göre ayrı bir değerler kümesi içeriyorsa buna ayrık zamanlı sinyal denir. Sonlu değerler kümesi içerir.



Ayrık Zamanlı Sinyallerin Gösterimi:



Grafiksel gösterim

$$x(n) = \begin{cases} 1, & \text{for } n = 1,3 \\ 4, & \text{for } n = 2 \\ 0, & \text{otherwise} \end{cases}$$

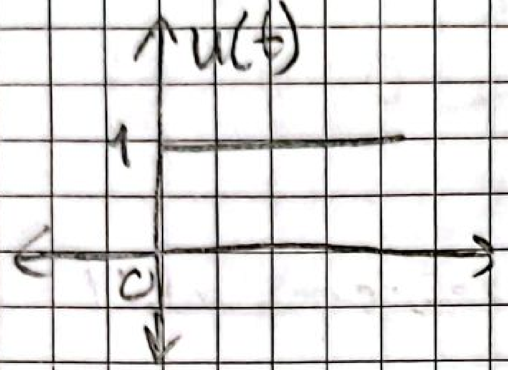
fonksiyon gösterim

n	...	-2	-1	0	1	2	3	4	...
x(n)	...	0	0	0	1	4	1	0	...

tablo gösterimi

Temel Sinyal Türleri

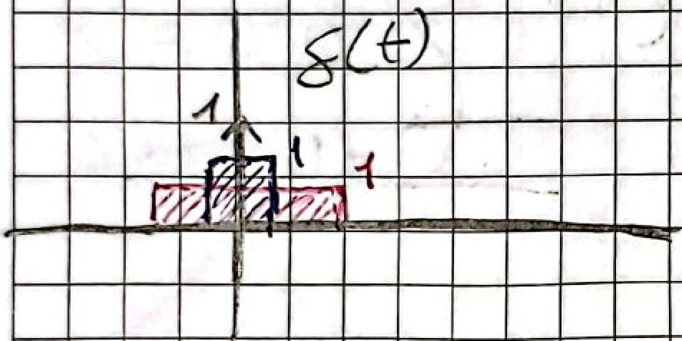
1- Birim Adım Fonksiyonu



$$u(t)=\begin{cases}1, & t\ge 0\\ 0, & t<0\end{cases}$$

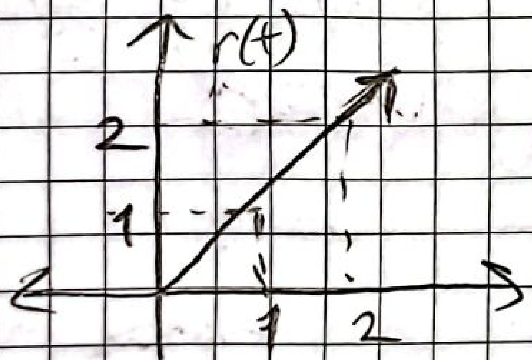
* En iyi test sinyalidir.

2- Birim İmpuls Fonksiyonu



$$\int_a^b \delta(t)\,dt = \begin{cases}1, & a<0<b\\ 0, & \text{otherwise}\end{cases}$$

3- Rampa Sinyali



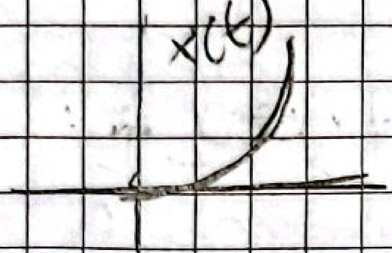
$$r(t)=\begin{cases}t, & t\ge 0\\ 0, & t<0\end{cases}$$

* $\int u(t)\,dt = \int 1\,dt = t = r(t)$

$$\Downarrow$$

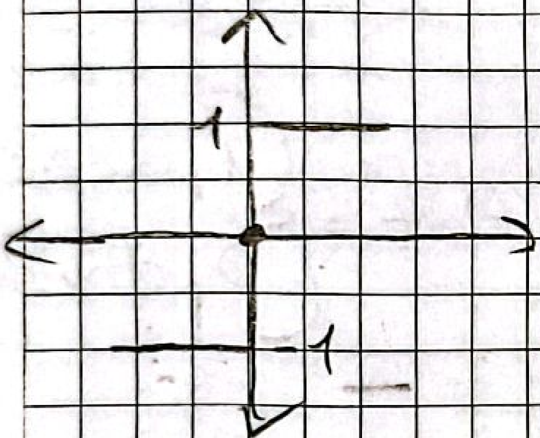
$$\boxed{u(t)=\frac{dr(t)}{dt}}$$

4- Parabolik Sinyal



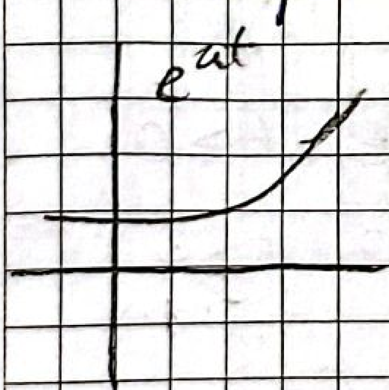
$$x(t) = \begin{cases} t^2/2, & t \geq 0 \\ 0, & t < 0 \end{cases}$$

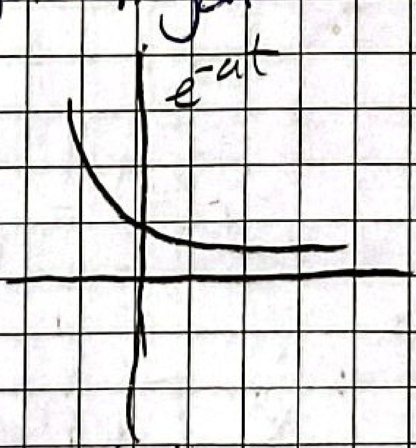
5- Signum Function



$$\mathrm{sgn}(t) = \begin{cases} 1, & t > 0 \\ 0, & t = 2 \\ -1, & t < 0 \end{cases}$$

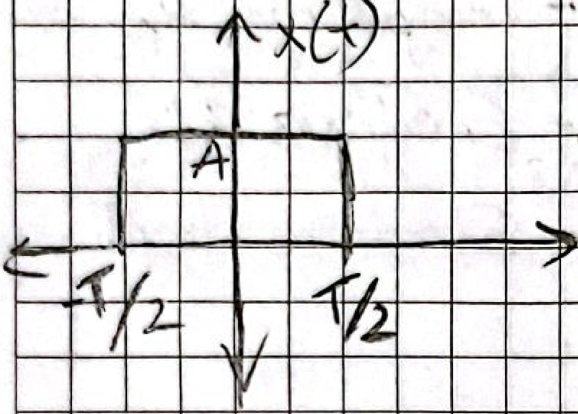
6- Eksponansiyel Sinyal





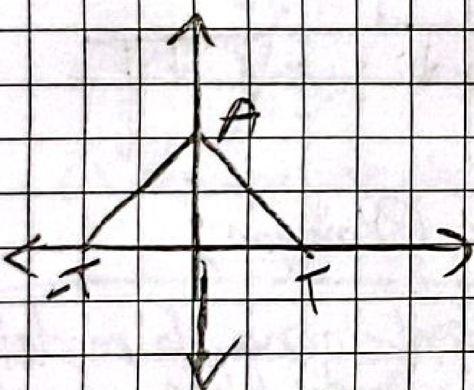
$$x(t) = e^{at}$$

7- Dortgensel Sinyal



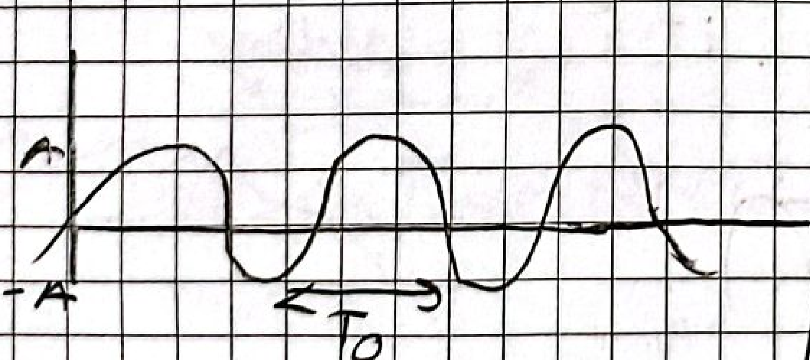
$$x(t) = A\,\mathrm{rect}\left[\frac{t}{T}\right]$$

8- Üçgensel Sinyal



$$x(t) = A\left[1-\frac{|t|}{T}\right]$$

9- Sinüzoidal Sinyal



$$x_1(t) = A\cos(w_0 \pm \phi)$$

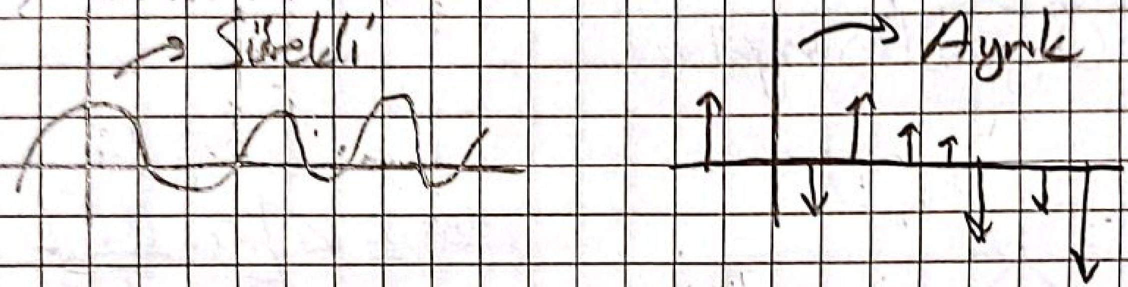
$$x_2(t) = A\sin(w_0 \pm \phi)$$

Where $T_0 = 2\pi / w_0$

Sinyaller aşağıdaki 6 sınıfta sınıflandırılabilir:

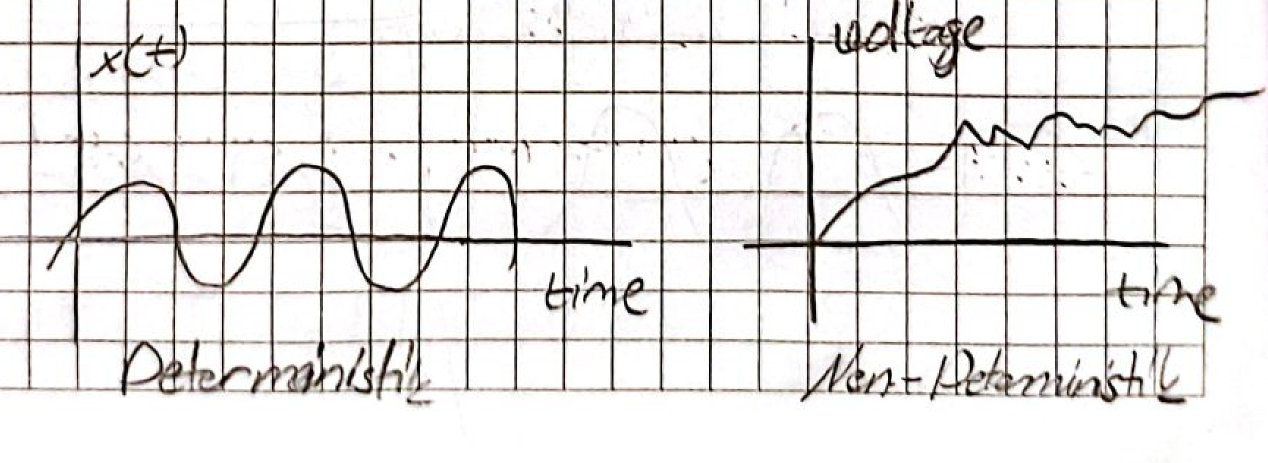
1- Sürekli ve Ayrık zamanlı
2- Kararlı ve Kararsız
3- Tek ve Çift
4- Periyodik ve Aperiyodik
5- Enerji ve Güç
6- Gerçek ve Hayali

1- Sürekli ve Ayrık Zamanlı;



2- Determenistik ve Determenistik Olmayan;

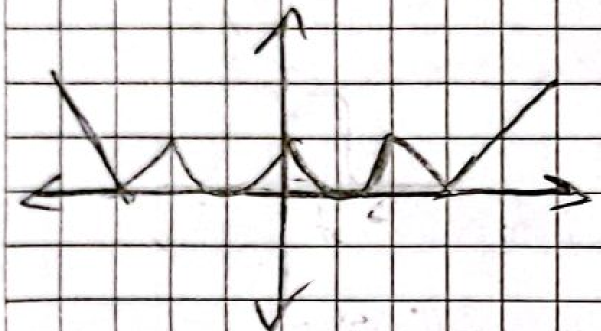
Determenistik sinyaller zamanın bir fonksiyonu ile modellenebilir (t). Rastgele sinyaller her hangi bir zamanda rastgele değerler alan ve istatiksel olarak karakterize edilemeyen sinyallerdir.



3- Tek ve Çift ;

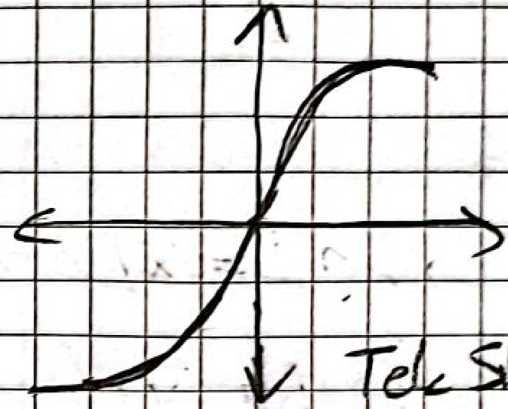
Eğer bir sinyal $x(t) = x(-t)$ koşulunu sağlarsa bu sinyal çift sinyaldir.

Eğer bir sinyal $x(t) = -x(-t)$ koşulunu sağlarsa bu sinyal tek sinyaldir.



Çift Sinyal

(x ekseninе göre simetrik)



Tek Sinyal

(orijine göre simetrik)

$$x(t) = x_e(t) + x_o(t)$$

$$x_e(t) = \frac{1}{2}[x(t) + x(-t)]$$

$$x_o(t) = \frac{1}{2}[x(t) - x(-t)]$$

$$x(t) = \frac{1}{2}[x(t) + x(-t)] + \frac{1}{2}[x(t) - x(-t)]$$

4- Periyodik ve Aperiyodik;

$x(t) = x(t+T)$ for all t,

T bir pozitif sabit değer olmak üzere, Eğer $T = T_0$ şartını sağlıyorsa tüm $T = 2T_0, 3T_0, 4T_0$ değerlerini de sağlar.

$f = 1/T \qquad w = 2\pi f = 2\pi/T$

Ayrık zamanlı için;

$x[n] = x[n+N]$ for integer n,

$f = 1/N \qquad \Omega = 2\pi/N$

Önerme $\rightarrow$ $x[k] = A\sin(\Omega_0 k + \theta)$

$\Omega_0/2\pi = m/k_0 \rightarrow k_0 = 2\pi m/\Omega_0$

Örnekler

1- $f[k] = \sin(\pi k/12 + \pi/4) \rightarrow k_0 = 2\pi m/\pi/12$

$\boxed{k_0 = 24m}$ $\boxed{m=1}$

2- $h[k] = \cos(0.5k + \phi) \rightarrow k_0 = 2\pi m/0.5$

$\boxed{k_0 = 4\pi m}$ $\rightarrow$ Aperiyodik

3- $p[k] = e^{j(7\pi k/8 + \theta)}$

$k_0 = 2\pi m/7\pi/8 \Rightarrow \boxed{k_0 = \frac{16m}{7}}$ $\boxed{m=7}$

5- Enerji ve Güç;

(1) Bir sinyalin gücü $\longrightarrow p(t) = x^2(t)$

(2) Bir sinyalin toplam enerjisi $\rightarrow E = \lim_{T\to\infty} \int_{-T/2}^{T/2} x^2(t)\,dt$

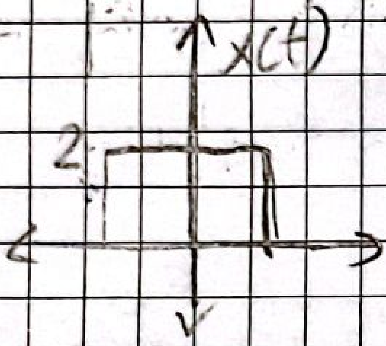
$$\star \; E = \int_{-\infty}^{\infty} x^2(t)\,dt \; \star$$

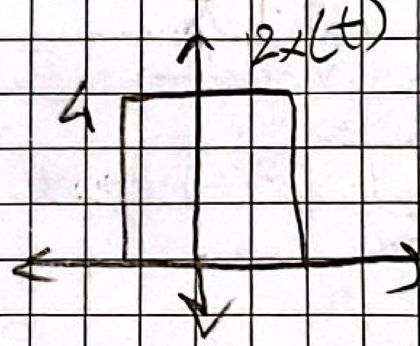
(3) Bir sinyalin ortalama gücü $\rightarrow P = \lim_{T\to\infty} \frac{1}{T} \int_{-T/2}^{T/2} x^2(t)\,dt$

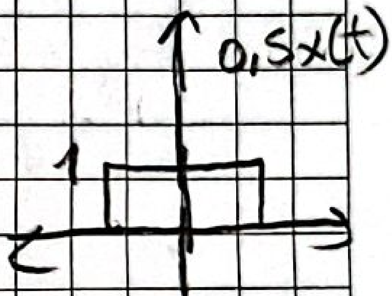
2

Sinyallerde Temel İşlemler

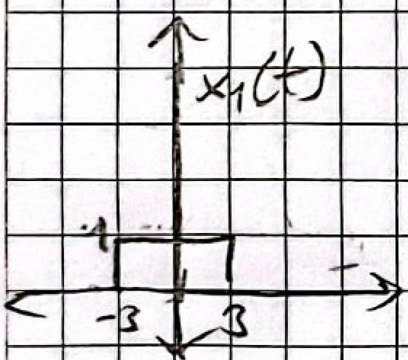
1- Genlik Ölçekleme

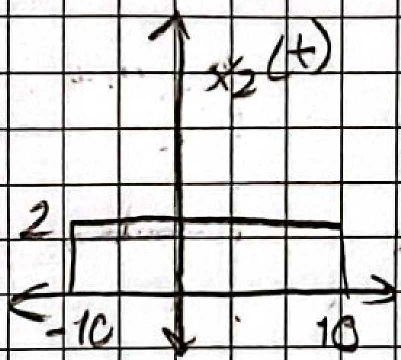




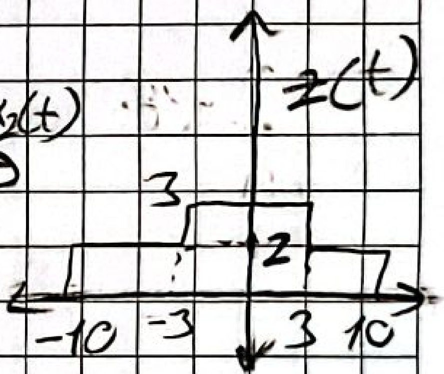


2- Ekleme (Toplama)

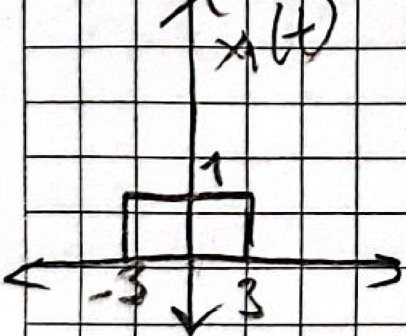


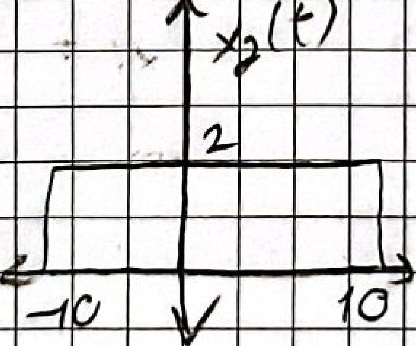


$\xrightarrow[z(t)]{x_1(t)+x_2(t)}$

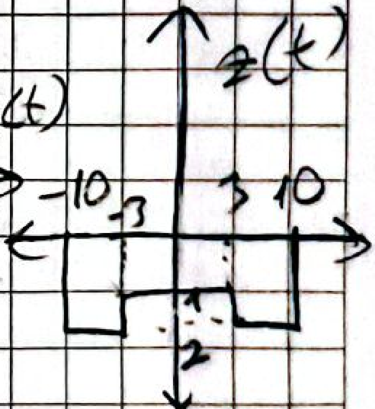


3- Çıkarma

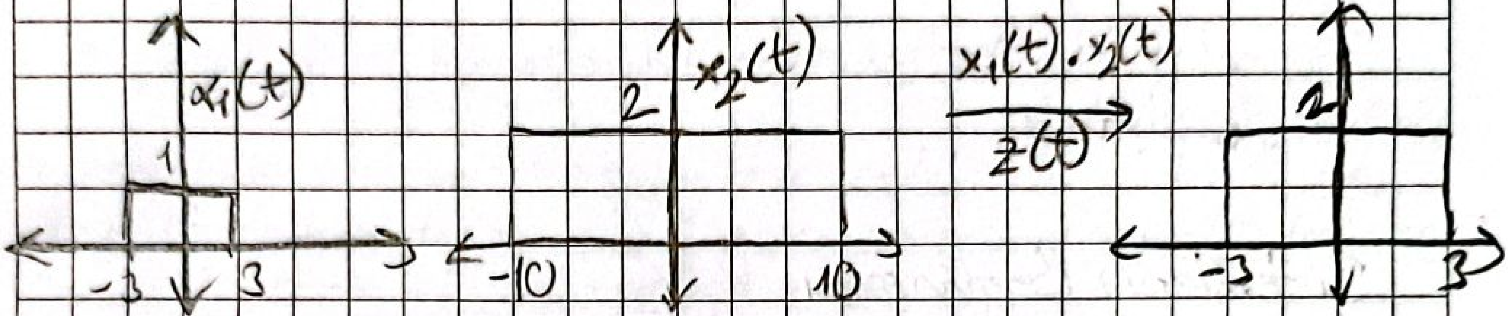




$\xrightarrow[z(t)]{x_1(t)-x_2(t)}$

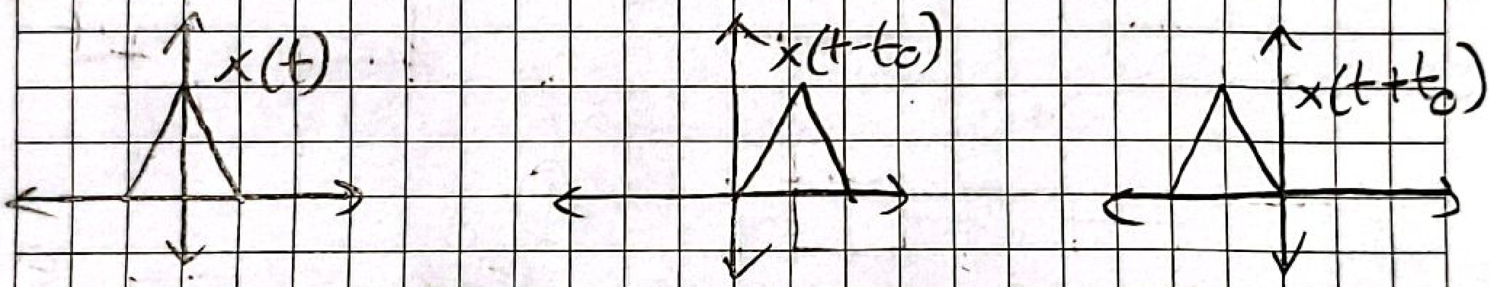


4- Çarpma

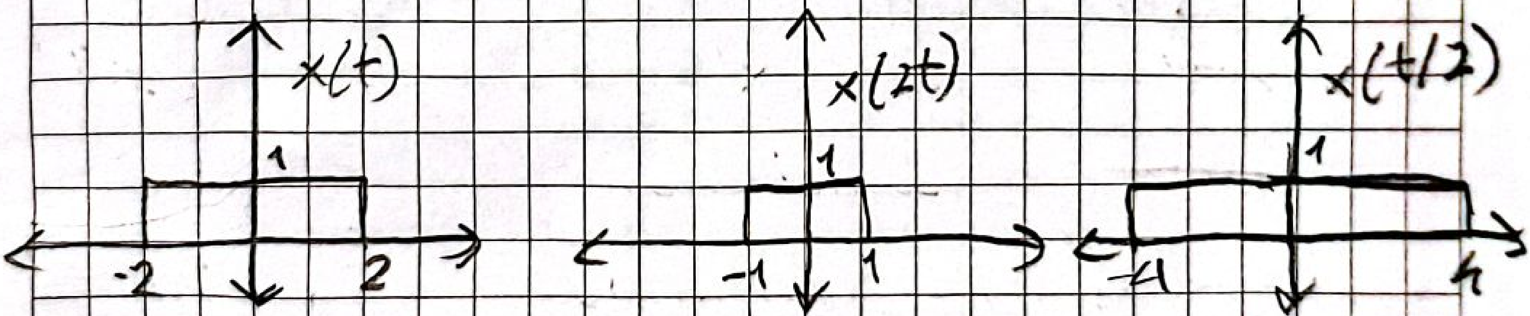


5- Time Shifting

$x(t \mp t_0)$, $x(t - t_0) \rightarrow$ artı yönde , $x(t + t_0) \rightarrow$ eksi yönde



6- Time Scaling



Öncelik Sıralaması $\rightarrow$ 1- Shifting 2- Scaling 3- Reversal

3

Sistemler

Fiziksel yada matematiksel bir sinyal üretir yada sinyal dönüştürür.

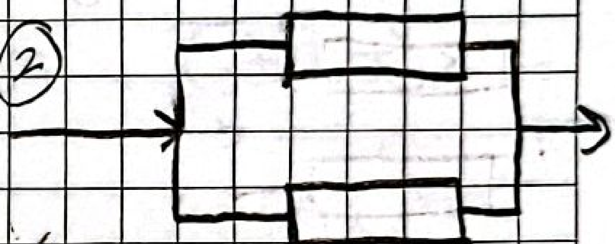
Sistemlerin Bağlanması

Sistemler seri, paralel, seri-paralel ve geri beslemeli olmak üzere dört farklı şekilde bağlanabilir.

1

Seri

2

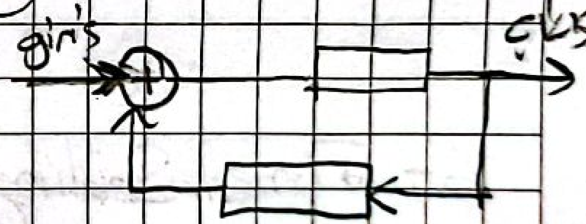


Paralel

3

Seri-Paralel

4



Geri Beslemeli

Sistemlerin Temel Özellikleri

1- Hafızalı ve Hafızasız Sistemler

Eğer bir sistemin herhangi bir andaki çıkışı, bağımsız değişkenin her değeri için girişin sadece o andaki değerine bağlı ise sistem hafızasızdır denir.

Örnek → $y[n] = (2x[n] - x[n])^2$

* Sadece o anki değere bağlı, sistem hafızasızdır.

Örnek → $y[n] = \sum_{-\infty}^{n} x[n]$

* Sistem çıkışı $x[n]$'nin önceki ve sonraki değerlerine de ihtiyaç duyar, sistem hafızalıdır.

2- Zamanla Değişen ve Zamanla Değişmez Sistemler

Eğer girişindeki zaman ötelemesi çıkışında da zaman ötelemesine sebep oluyorsa bu sisteme zamanla değişmez sistem denir.

$x(t) \rightarrow \boxed{\text{sistem}} \rightarrow y(t)$

$y(t) = \sin(x(t))$

$y_1(t) = \sin(x_1(t))$

$x_2(t) = x_1(t - t_0)$

$y_2(t) = \sin(x_2(t)) = \sin(x_1(t - t_0))$

Devam → $y_1(t - t_0) = \sin(x_1(t - t_0))$

$y_2(t) = y_1(t - t_0)$

Zamanla Değişmez Sistem

3- Geri Dönüştürülebilirlik ve Ters Sistemler

Bir sistemde eğer farklı girişler farklı çıkışlara neden oluyorsa geri dönüştürülebilirdenir.

$y(t) = 5x(t)$

$x(t) = 3 \rightarrow y(t) = 15$

$x(t) = -3 \rightarrow y(t) = -15$

Terslenebilir ✓

$y(t) = x^2(t)$

$x(t) = 3 \rightarrow y(t) = 9$

$x(t) = -3 \rightarrow y(t) = -9$

Terslenemez ✗

Sistemlerin Temel Özellikleri

1- Nedensellik (Causality)

Bir sistemin t ya da n anındaki çıkışı, geçmiş ya da o andaki giriş değerine bağlı ise sistem nedenseldir. Eğer sistem gelecek giriş değerlerine de bağlı ise nedensel değildir.

Örnek $\rightarrow y[n] = 0,2 \cdot (x[n-3] + x[n-1] + x[n])$

Nedensel sistem

Örnek $\rightarrow y[n] = 0,2 \cdot (x[n-2] + x[n] + x[n+1])$

Nedensel olmayan sistem

2- Kararlılık (Stability)

Küçük girişlerin küçük çıkışlar oluşturduğu sistemlere kararlı sistemler denir. Başka bir ifade ile kararlı bir sistemde sınırlı girişler sınırlı çıkışlar oluşturur.

Örnek → $y[n] = 1/3 \left(x[n] + x[n-1] + x[n-2]\right)$

Kararlıdır çünkü küçük girdilerden çıktı üretir.

Örnek → $y[n] = r^n x[n]$, $r > 1$

Kararsızdır, r^n ifadesi yüzünden büyük çıktı üretir.

3- Doğrusallık (Linearity)

Bir sürekli zaman sisteminin $x_1(t)$'ye cevabı $y_1(t)$ ve $x_2(t)$'ye cevabı $y_2(t)$ olsun. Eğer aşağıdaki iki koşul sağlanıyorsa sistem lineerdir denir;

1- Sistemin "$x_1(t) + x_2(t)$"ye cevabı "$y_1(t) + y_2(t)$"dir.

"Toplamsallık"

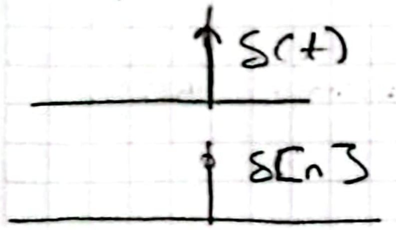
2- Sistemin "$a.x_1(t)$"ye cevabı "$a.y_1(t)$"dir.

"Çarpımsallık"

Genel gösterim → $a.x_1(t) + b.x_2(t) = a.y_1(t) + b.y_2(t)$

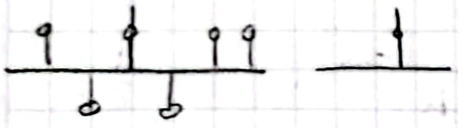
Time Domain Representation of Linear Time-Invariant System (LTI)

analog → continous system
computer → discrete / digital signal



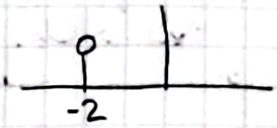
· impulse response ⇒ δ → input
· The convolutional Sum
$x[n]\,\delta[n] = x[0]\,\delta[n]$

δ'yı kaydırarak x[n]'in her durumu



$x[n]\,\delta[n-k] = x[k]\,\delta[n-k]$

$\delta[n+2]$



$x[n] = \dots\; x[-2]\,\delta[n+2] + x[-1]\,\delta[n+1] + x[0]\,\delta[n] + x[1]\,\delta[n-1] + x[2]\,\delta[n-2] \dots$

$$x[n] = \sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]$$

$x[n] \rightarrow \boxed{H} \rightarrow y[n]$ $\quad y[n] = H\{x[n]\}$ → sistemin transfer fonksiyonu

$$y[n] = H\left\{\sum_{k=-\infty}^{\infty} x[k]\,\delta[n-k]\right\} \Rightarrow y[n] = \sum_{k=-\infty}^{\infty} H\{x[k]\,\delta[n-k]\}$$

ikisi de lineer.

$$\Rightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]\,H\{\delta[n-k]\} \Rightarrow y[n] = \sum_{k=-\infty}^{\infty} x[k]\,h[n-k]$$ Konvolüsyon

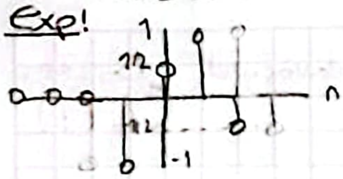
response $h[n-k]$ · kaymış impulse response'ların ağırlıklandırılmış toplamı

$\Rightarrow y[n] = x[n] * h[n]$
↳ konvolüsyon sembolü

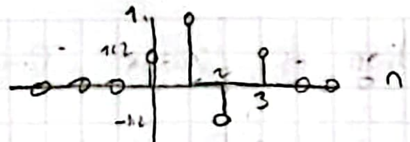
· ayrık zaman | convolutional sum
· continous " integral

Exp!

$x[n] = x[k]$



$h[n] = h[k]$

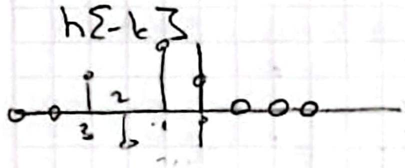


$y[n] = ?$ Konvolüsyon

$$y[n] = x[n] * h[n] = \sum_{k=-\infty}^{\infty} x[k]\, \delta[n-k]$$

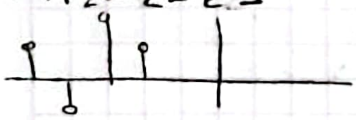
→ h'ı ters'e
→ impulse response'u en sola çek ve kaydırmaya başla

$h[-k]$



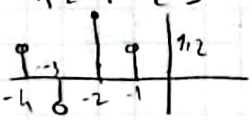
-2 -1 0 1 2

• $h[-2-k]$



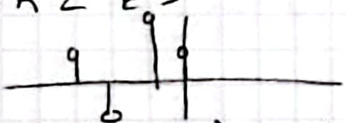
$y[-2] = 0$

• $h[-1-k]$



$y[-1] = (-1)\left(\frac{1}{2}\right) = -\frac{1}{2}$

• $h[-k]$

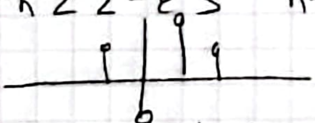


$y[0] = (-1).1 + \frac{1}{2}.\frac{1}{2} = -1 + \frac{1}{4} = \frac{3}{4}$

• $h[1-k]$

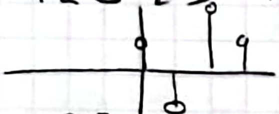
$y[1] = -1.\left(\frac{-1}{2}\right) + \frac{1}{2}.1 + 1 \times \frac{1}{2} = \frac{3}{2}$

• $h[2-k]$ $n=2$



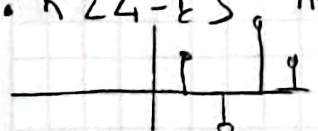
$y[2] = -1.\left(\frac{1}{2}\right) + \frac{1}{2}\times\left(\frac{-1}{2}\right) + 1.1 + \left(\frac{-1}{2}\right).\frac{1}{2}$
$= -\frac{1}{2} - \frac{1}{4} + 1 - \frac{1}{4} = 0$

• $h[3-k]$ $n=3$



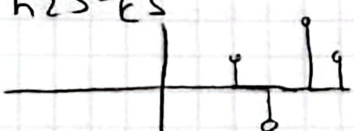
$y[3] = \frac{1}{2}.\frac{1}{2} - \frac{1}{2}.1 - \frac{1}{2}.1$
$= -1 + \frac{1}{4} = -\frac{3}{4}$

• $h[4-k]$ $n=4$



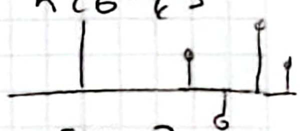
$y[4] = \frac{1}{2}.1 - \frac{1}{2}.\left(\frac{-1}{2}\right) + 1.0$
$= \frac{1}{2} + \frac{1}{4} = \frac{3}{4}$

• $h[5-k]$



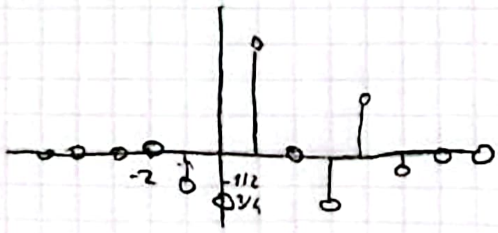
$y[5] = \frac{1}{2}.\left(\frac{-1}{2}\right) = -\frac{1}{4}$

• $h[6-k]$



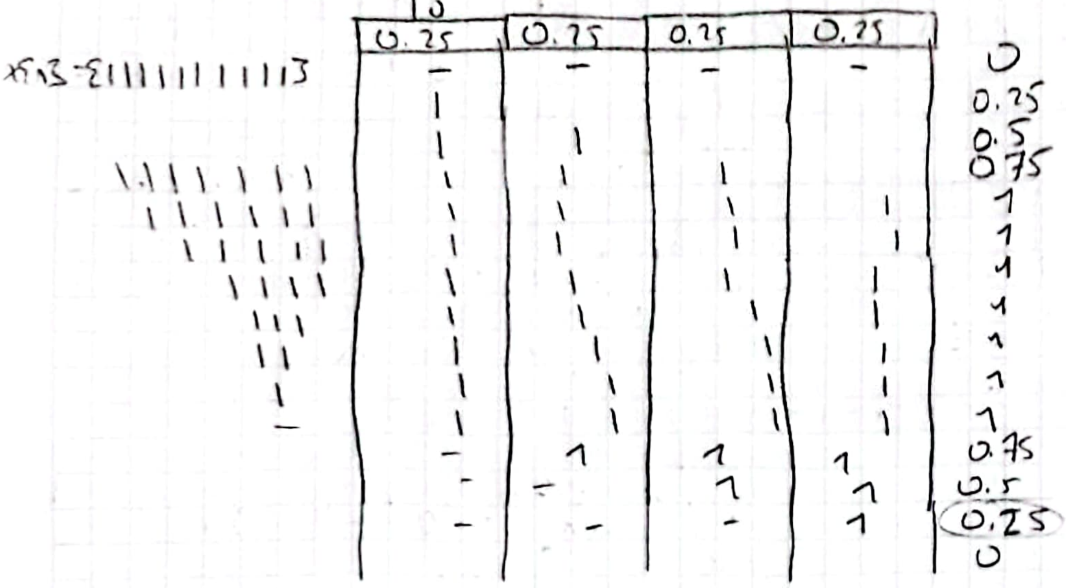
$y[6] = 0$

$y[-2] = 0$
$y[-1] = -1/2$
$y[0] = -3/4$
$y[1] = 3/2$
$y[2] = 0$
$y[3] = -3/4$
$y[4] = 3/4$
$y[5] = -1/4$
$y[6] = 0$



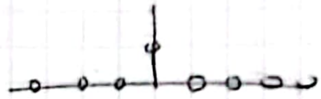
exp: $h[n]: \{\underset{\uparrow 0}{0.25} \quad 0.25 \quad 0.25 \quad 0.25\}$

$x[n]: \{\underset{\uparrow 0}{1} \; 1 \; 1 \; 1 \; 1 \; 1 \; 1 \; 1 \; 1 \; 1\}$



$y[n] = \{\underset{\uparrow 0}{0}\; .25 \quad 0.5 \quad 0.75 \quad 1 \quad 1 \; 1 \quad 1 \quad 1 \quad 1 \; 1 \quad 0.75 \quad 0.5 \quad 0.25\}$

exp: $y[n] = x[n] + \frac{1}{2} x[n-1]$



$$x[n] = \begin{cases} 2 & n=0 \\ 4 & n=1 \\ -2 & n=2 \\ 0 & \text{otherwise} \end{cases}$$

$x[n] = \delta[n]$

$y[n] = h[n]$

$h[n] = \delta[n] + \frac{1}{2}\delta[n-1]$

$h[-2] = \delta[-2] + \frac{1}{2}\delta[-3] = 0$

$h[1] = \frac{1}{2}\delta[0] = \frac{1}{2}$

$h[0] = 1$

$x[n] = 2\delta[n] + 4\delta[n-1] - 2\delta[n-2]$

$y[n] = 2h[n] + 4h[n-1] - 2h[n-2]$

$$= 2\left(\delta[n] + \frac{1}{2}\delta[n-1]\right) + 4\left(\delta[n-1] + \frac{1}{2}\delta[n-2]\right) - 2\left(\delta[n-2] + \frac{1}{2}\delta[n-3]\right)$$

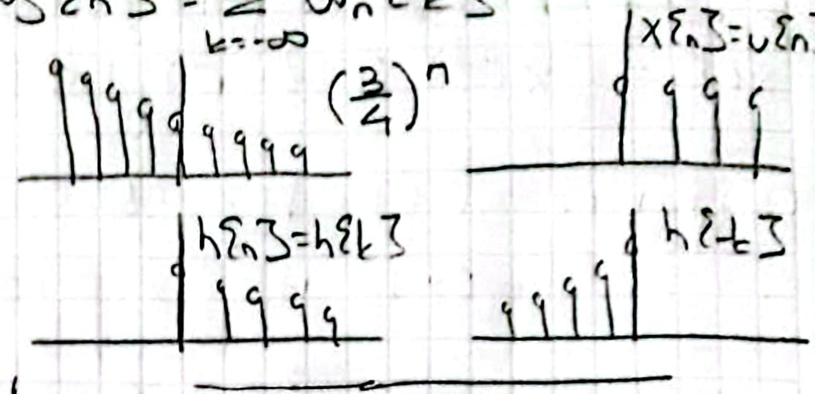
$= 2\delta[n] + \delta[n-1] + 4\delta[n-1] + 2\delta[n-2] - 2\delta[n-2] - \delta[n-3]$

$= 2\delta[n] + 5\delta[n-1] - \delta[n-3]$

$$y[n] = \begin{cases} 2 & n=0 \\ 5 & n=1 \\ 0 & n=2 \\ -1 & n=3 \\ 0 & \text{otherwise} \end{cases}$$

Singaller ve Sistemler - Convolutional Sum Evaluation Procedure

$$y[n] = \sum_{k=-\infty}^{\infty} \underbrace{x[k]h[n-k]}_{w_n[n]} \qquad y[n] = \sum_{k=-\infty}^{\infty} w_n[k]$$



<u>ex:</u> $h[n] = \left(\frac{3}{4}\right)^n u[n]$

$x[n] = u[n]$

$$y[n] = \sum_{k=-\infty}^{\infty} x[k]h[n-k]$$

$$y[n] = \sum_{k=-\infty}^{\infty} u[n]h[n-k]$$

$$= \sum_{k=-\infty}^{\infty} u[n] \cdot \left(\frac{3}{4}\right)^{n-k} u[n-k]$$

$$= \sum_{k=0}^{\infty} \left(\frac{3}{4}\right)^{n-k} u[n-k] \qquad n-k \geq 0$$

$$y[n] = \sum_{k=0}^{n} \left(\frac{3}{4}\right)^{n-k}$$

$y[5] = 3.280$

$$y[-5] = \sum_{k=0}^{-5} \dots = 0$$

$n = 5$

$$y[5] = \sum_{k=0}^{5} \left(\frac{3}{4}\right)^{5-k} = \sum_{k=0}^{5} \left(\frac{3}{4}\right)^{5} \cdot \left(\frac{4}{3}\right)^k$$

$$= \left(\frac{3}{4}\right)^5 \sum_{k=0}^{5} \left(\frac{4}{3}\right)^k = \left(\frac{3}{4}\right)^5 \cdot \left(\frac{1-\left(\frac{4}{3}\right)^6}{1-\frac{4}{3}}\right)$$

$$= 0.237 \cdot \left(\frac{1-5.61}{-0.333}\right) \qquad s = \sum_{k=0}^{n} r^k = \frac{1-r^{n+1}}{1-r}$$

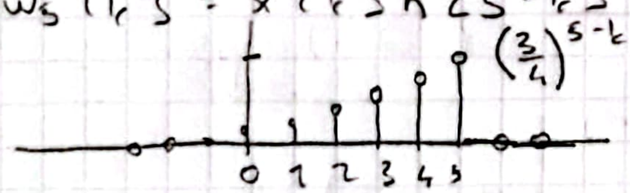
$$= 0.237 \cdot \left(\frac{-4.61}{-0.333}\right) = 3.280 \qquad s = \sum_{k=0}^{\infty} r^k = \frac{1}{1-r}$$

$$y[10] = \sum_{k=0}^{10} \left(\frac{3}{4}\right)^{10-k} = \left(\frac{3}{4}\right)^{10} \sum_{k=0}^{10} \left(\frac{4}{3}\right)^k = (0.05)\,\frac{1-\left(\frac{4}{3}\right)^{11}}{1-4/3}$$

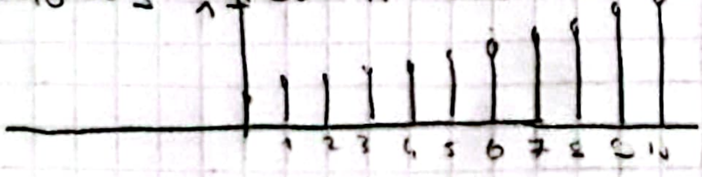
$$= (0.05) \cdot \frac{1-23.67}{-0.333} = 3.832$$

$w_{-5}[k] = x[k]h[-5-k]$

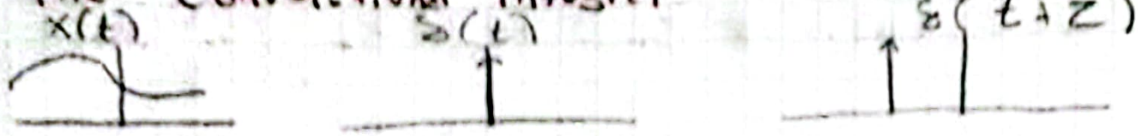
$w_5[k] = x[k]h[5-k]$



$w_{10}[k] = x[k]h[10-k]$



The Convolutional Integral



$$x(t) = \int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau$$

Output $\Rightarrow y(t) = H\{x(t)\}$

$$y(t) = H\left\{\int_{-\infty}^{\infty} x(\tau)\,\delta(t-\tau)\,d\tau\right\}$$

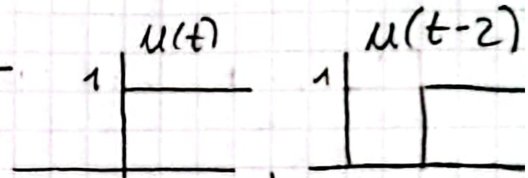
$$= \int_{-\infty}^{\infty} x(\tau)\,\underbrace{H\{\delta(t-\tau)\}}_{h(t-\tau)}\,d\tau$$

$$\boxed{y(t) = x(t) * h(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau}$$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$$

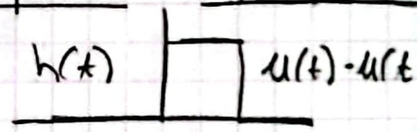
$$w_t(\tau) = x(\tau)\,h(t-\tau)$$

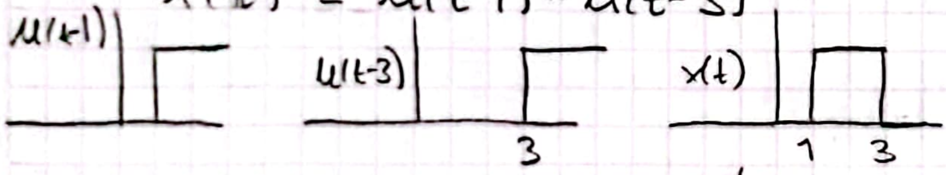
$$y(t) = \int_{-\infty}^{\infty} w_t(\tau)\,d\tau$$



Exp: $h(t) = u(t) - u(t-2)$

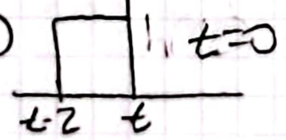
$x(t) = u(t-1) - u(t-3)$





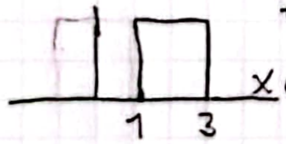
$u(t-1) - u(t-3)$

$\cdot\, h(t) \to h(\tau) \to h(-\tau)$



t=0 kesisim yok

$\cdot\, x(t) \to x(\tau)$



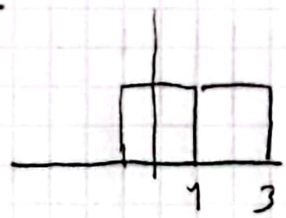
$x(t) = x(\tau)$

$$y(t) = \int_{-\infty}^{\infty} x(\tau)\,h(t-\tau)\,d\tau$$

t=1 olunca kesisin basla.

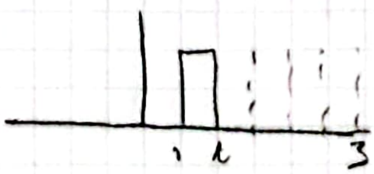
$$= \int_{1}^{t} 1.1\,d\tau = \tau\Big|_1^t$$

$$= t-1$$



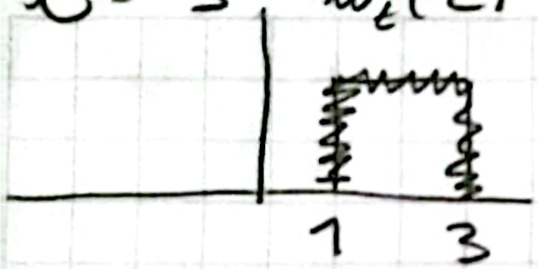
$1 \le t \le 3$

$t-1$

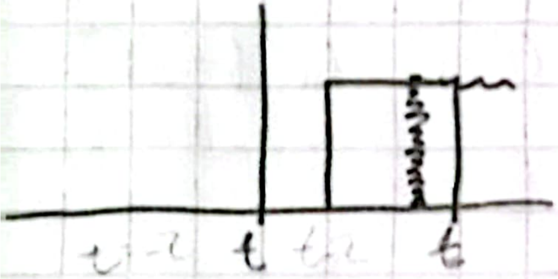


$w_t(\tau)$

$t=3$



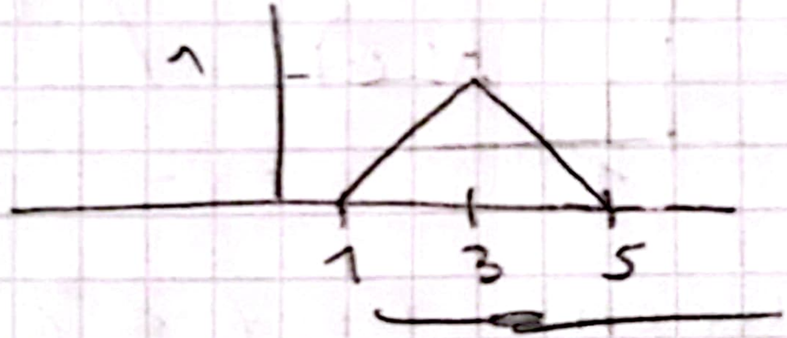
$$y(t)=\int_1^3 1.1\,d\tau = \tau\Big|_1^3 = 2$$



$$y(t)=\int_{t-2}^{3} x(\tau)\,h(t-\tau)\,d\tau \qquad 3\le t\le 5$$

$$=\int_{t-2}^{3} 1.1\,d\tau = \tau\Big|_{t-2}^{3} = 3-t+2$$

$$=5-t$$



<u>exp</u>: $h(t)=u(t)-u(t-2)$

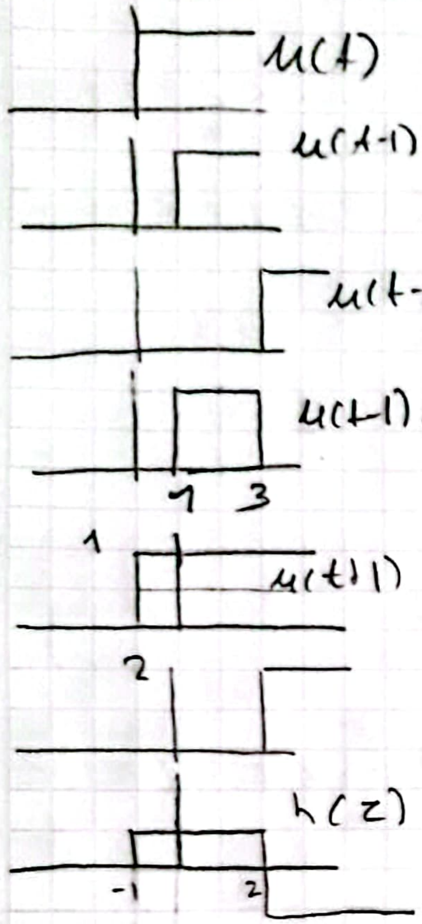
$x(t)=u(t-1)-u(t-3)$

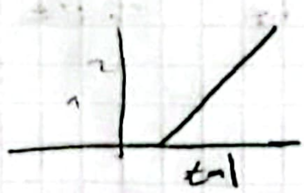
$h(t)=u(t)-u(t-2)$

$$y(t)=\int_{-\infty}^{\infty} x(\tau)\,h(t$$

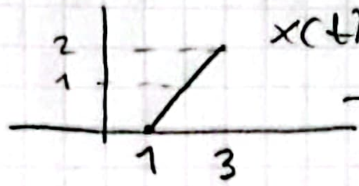
exp: $x(t) = (t-1)[u(t-1) - u(t-3)]$

$h(t) = u(t+1) - 2u(t-2)$

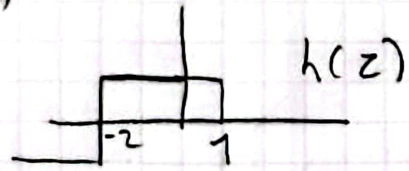




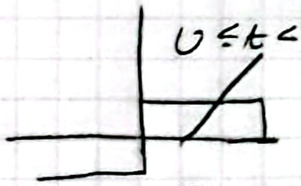
$x(t) = (t-1)[u(t-1) - u(t-3)]$



$\to (z-1)$



$\} \; t=0$



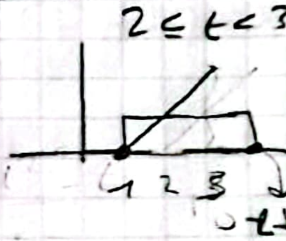
between $0 \leq t < 2$

$y(t) = \int_{-\infty}^{\infty} x(z)\, h(t-z)\, dz$

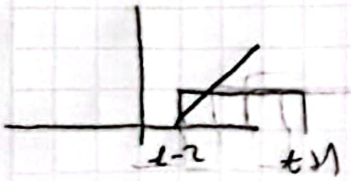
$= \int_{1}^{t+1} (z-1)\, 1\, dz = \frac{z^2}{2} - z \Big|_{1}^{t+1}$

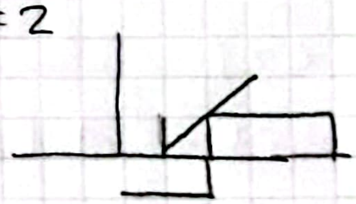
$= \frac{(t+1)^2}{2} - (t+1) - \frac{1}{2} + 1 = \frac{t^2}{2} + t + \frac{1}{2} - t - 1 + 1 - \frac{1}{2}$

$y(t) = t^2/2$



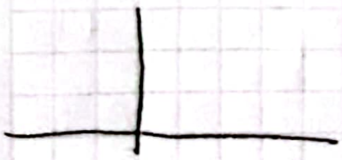
$y(t) = \int_{1}^{3} (z-1)\, dz = \frac{z^2}{2} - z \Big|_{1}^{3} = \frac{9}{2} - \frac{1}{2} - 3 + 1 = 2$

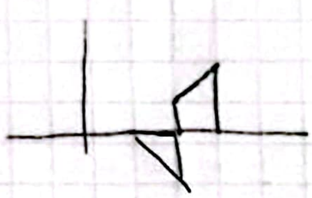




$y(t) = \int_{t-2}^{3} (z-1)\, 1\, dz + \int_{1}^{t-2} (z-1)\, 1\, dz$

$= -t^2 + 6t - 7$





devamı aşağıda var.